

Lecture 24

ChE 467/596-012 **Polymer Rheology**

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Moving on to Chap 11

Last topic:

Colloids and Suspensions

Today:

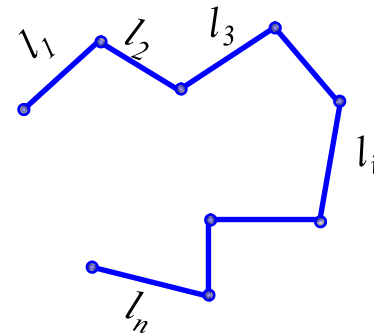
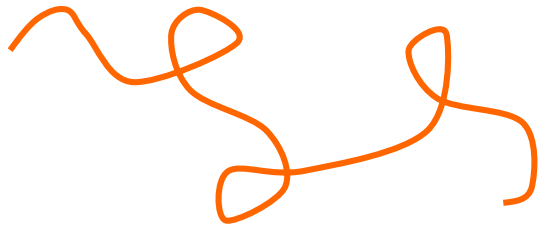
Revisit polymer

Conformation & chain dynamics

Models (physically based) & their predictions

Polymers

General definition & Concepts:

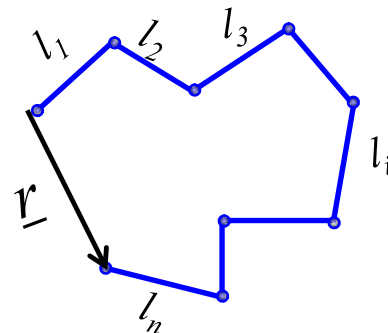


Consider polymer as a freely jointed individual segments

End to end distance:

$$\underline{r} = \sum \underline{l}_i$$

$$\underline{r}^2 = \underline{r} \cdot \underline{r} = \sum_i \sum_j \underline{l}_i \cdot \underline{l}_j$$



Polymers, cont'd

Freely jointed chains/segments:

$$\overline{r^2} = \sum_i \overline{l_i \cdot l_j} = n \overline{l^2}$$

mean square end to end distance *# of segments* *average length of each segment*

If not freely jointed:

$$\overline{r^2} = nl^2 \frac{(1 - \cos \theta)}{(1 + \cos \theta)}$$

For $\theta = 109.5^\circ$ (tetrahedral) $\overline{r^2} = 2nl^2$

Polymers, cont'd

$\bar{r}^2 = nl^2$, Completely Freely jointed chains (θ unrestricted)

$$\bar{r}^2 = nl^2 \frac{(1 - \cos \theta)}{(1 + \cos \theta)}, \quad \text{restricted } \theta$$

Assumptions:

- *No effect of solvent*
- *No effect of external force*

Either we are looking at melts or polymers in θ (theta) solvents

Polymers, cont'd

For synthetic polymers, can define characteristic ratio:

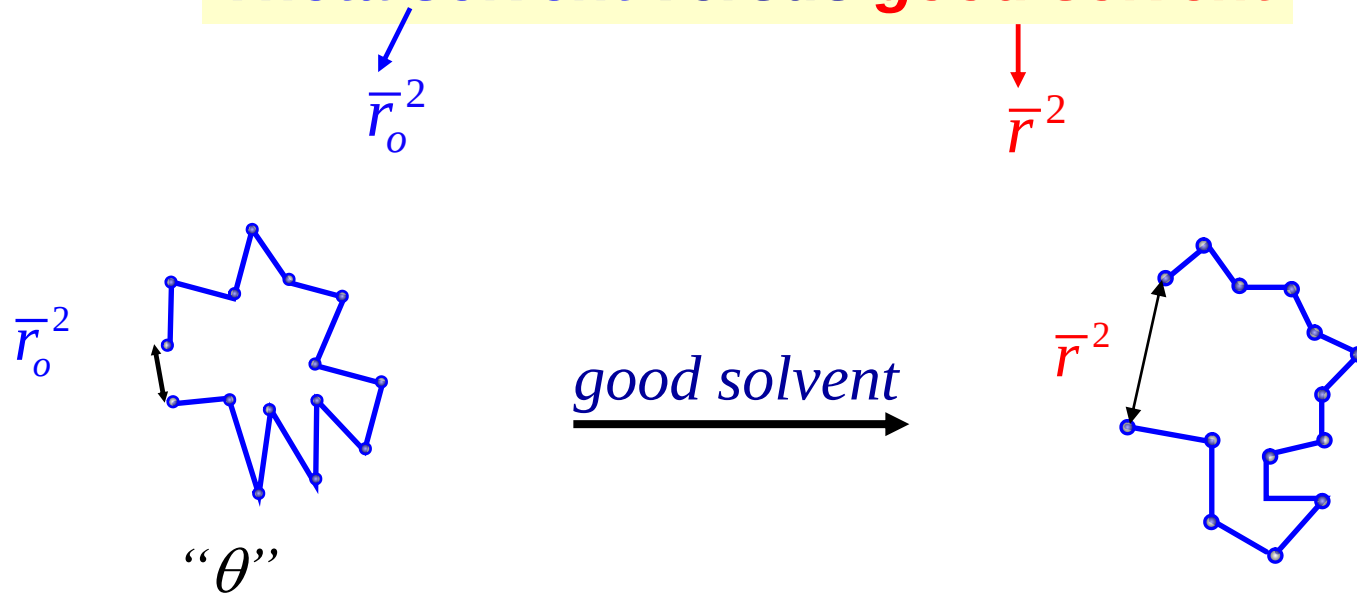
$$C \equiv \frac{\bar{r}_o^2}{nl^2}$$

$\bar{r}_o^2$ is the end to end distance at θ condition
(unperturbed chains)

- *C is a function of chain stiffness*
- *C increases with chain stiffness*
- *C ~ 5-10 for most synthetic polymers*

Polymers, cont'd

Theta solvent versus good solvent

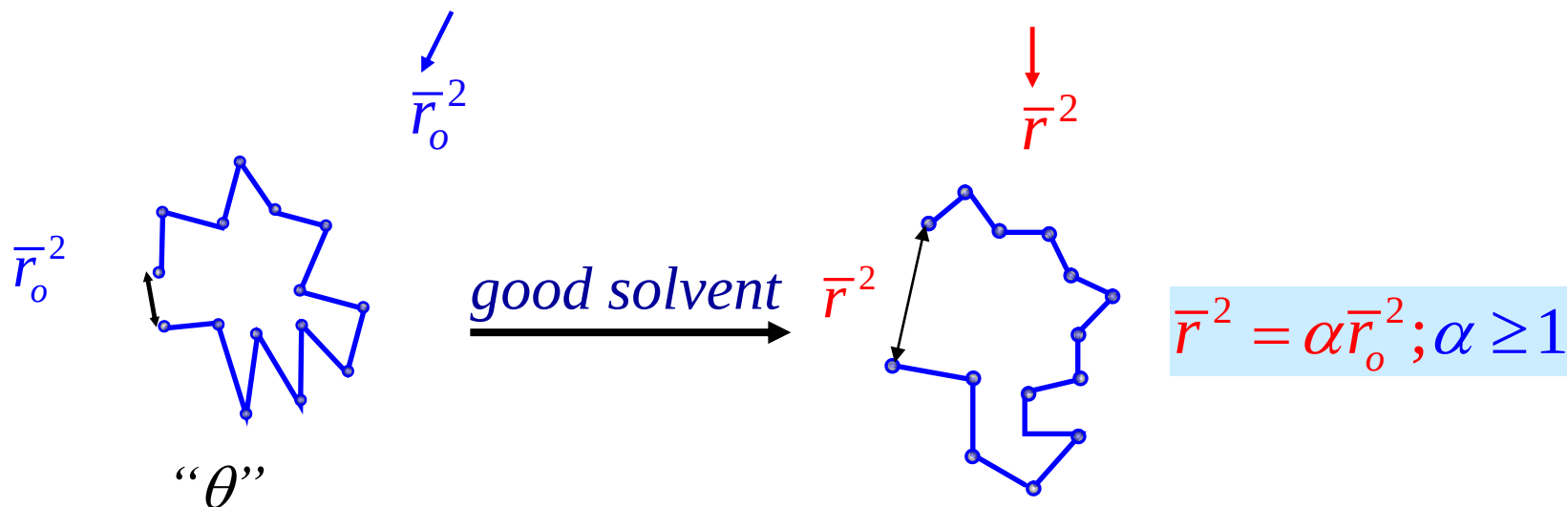


$$\bar{r}^2 = \alpha \bar{r}_o^2; \quad \alpha \geq 1$$

Polymer chain expands in a good solvent

In class problem

Theta solvent versus *good solvent*



Is $\bar{r}^2$ dependent on molecular weight?

Ans: *Yes*

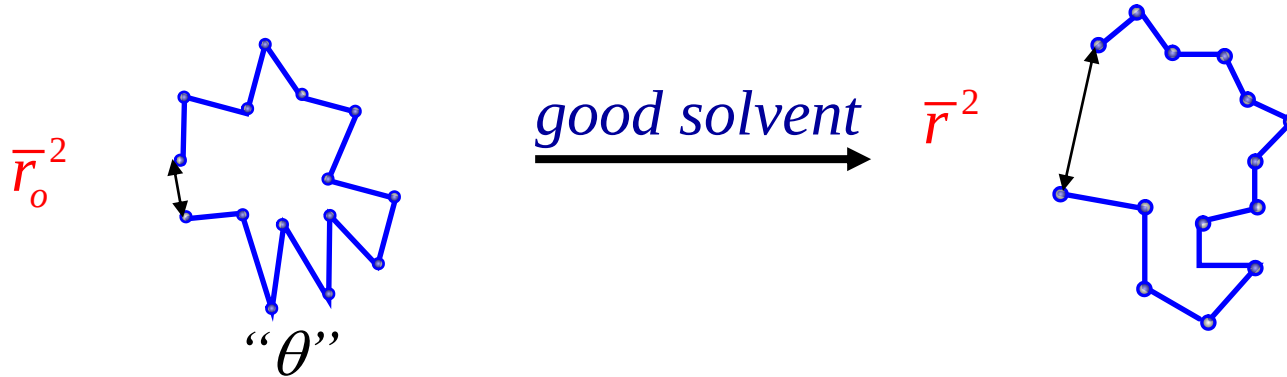
$$\bar{r}^2 = nl^2 = (nl)l$$

$$nl \sim \text{mol.wt.}(M)$$

Polymers, cont'd

Theta solvent versus good solvent:

$$\bar{r}^2 = \alpha \bar{r}_o^2; \quad \alpha \geq 1$$



$$\alpha^5 - \alpha^3 = A \left(1 - \frac{T_o}{T} \right) M^{1/2}$$

At $T = T_o$, $\alpha = ?$

At $T > T_o$, $\alpha > 1$ and depends on M.W. at most to the 1/10 power

An Issue!

For synthetic polymers, can define characteristic ratio:

$$C \equiv \frac{\bar{r}_o^2}{nl^2} \quad (C_\infty \text{ used instead of } C \text{ sometimes})$$

$\bar{r}_o^2$ is the end to end distance at θ condition
(unperturbed chains)

- *C increases with chain stiffness*
- *C ~ 5-10 for most synthetic polymers*

Issue: Since $\bar{r}^2 = nl^2$ and $\bar{r}^2 = \alpha \bar{r}_o^2$; $\alpha \geq 1$, how can C be 5-10?

$\bar{r}^2 = nl^2$ holds only for freely jointed chains

Real chains not freely jointed  $\bar{r}_o^2 > nl^2$

Some typical Values for C

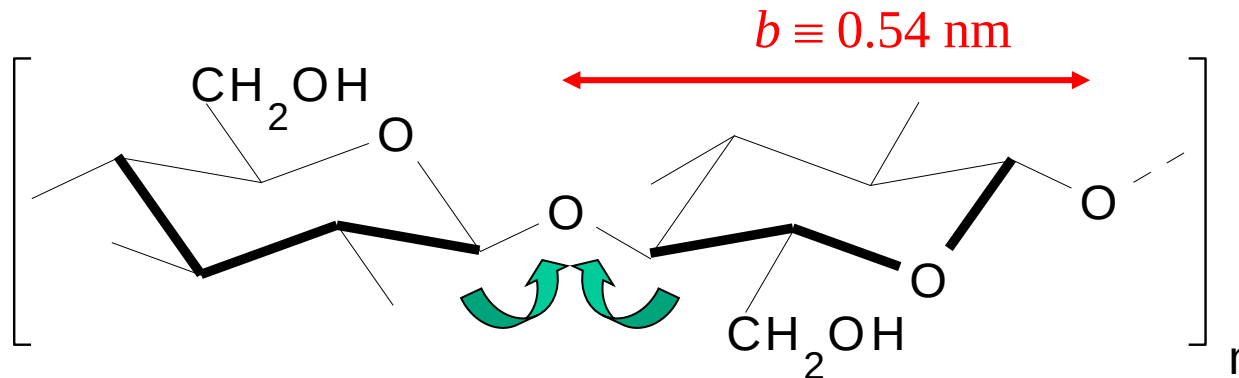
Polymer	C_{∞}	
Carboxymethyl amylose	5.3 - 10	α (1→4)
Carboxymethyl cellulose	14.2	β (1→4)
Guar galactomannan	12.6	β (1→4)
PVP	12.3	
PVOH (88:12)	8.3	
Poly (ethylene oxide)	4.0	

Expansion of chains

Chain dimensions expanded by imposition of fixed valance angles.

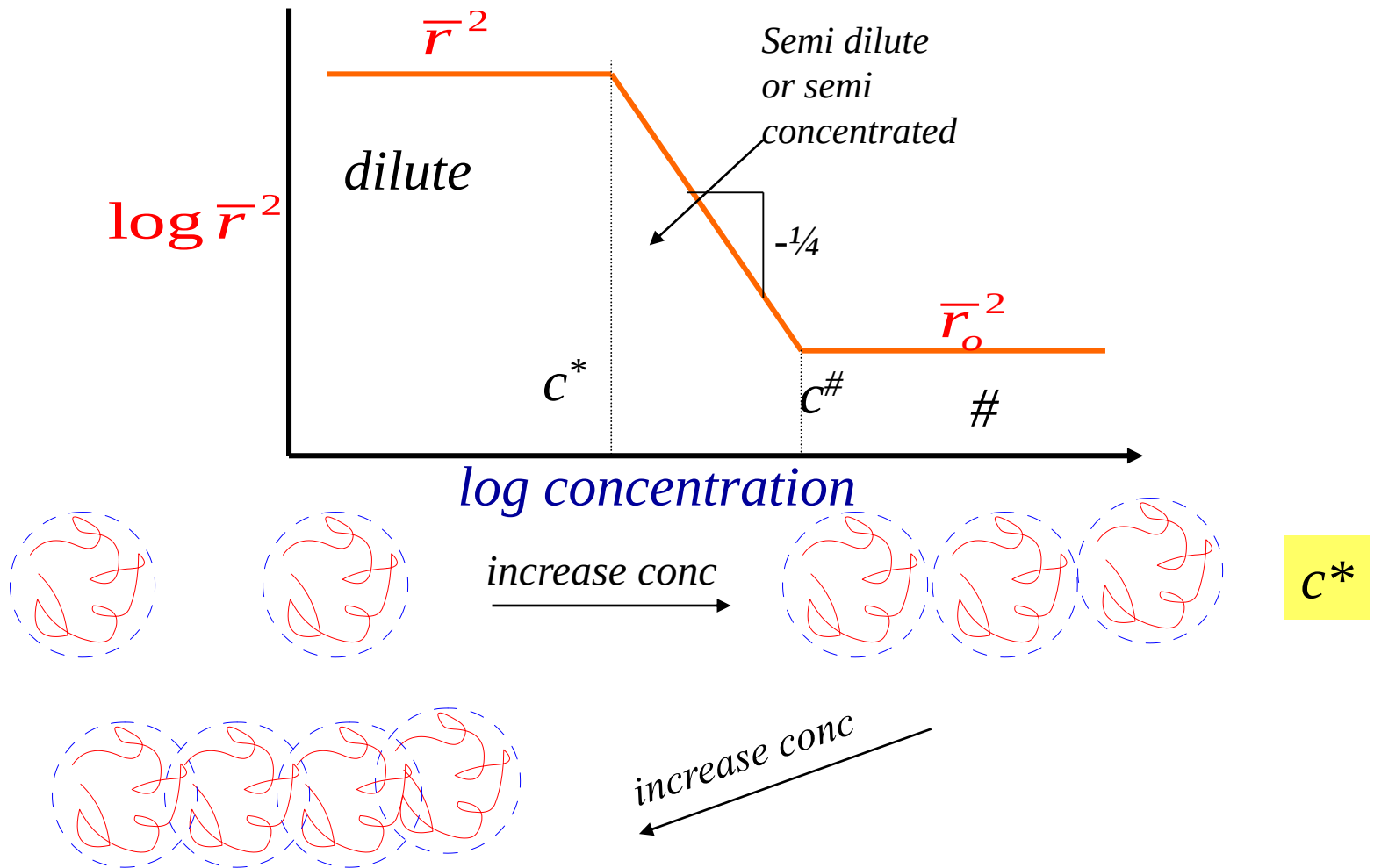
Also, steric repulsion between substituents on the backbone, renders many conformations energetically unfavourable, further expanding the chain dimensions

(1→4) - β - D - glucan



Steric clashes results in many of the dihedral angles obtained by rotational of the O-C bonds of the glycosidic linkage being precluded.

Polymer Concentration Regimes



Recap

$\bar{r}^2$ is the m.s end to end distance in arbitrary solvent

$\bar{r}_o^2$ is the end to end distance at θ condition

c^* - *critical overlap concentration*

$c^\#$ - *for $c > c^*$ [concentrated]*

R_g – radius of gyration

- the r.m.s. distance of elements of the chain from its centre of mass.

Zero Shear Viscosity

Dilute Solutions ($c < c^$)*

Einstein's equation for suspension

$$\eta = \eta_s (1 + 2.5\phi)$$

$$\frac{\eta - \eta_s}{\eta_s} = 2.5\phi = 2.5 \left(\frac{n_2}{V} \right) V_{\text{sphere}}$$

*#of molecules
volume of solvent*

V_{sphere} = vol of molecular spheres

$$V_{\text{sphere}} \text{ (for random coil)} = k_1 (\bar{r}^2)^{3/2}$$

$$\left(\frac{n_2}{V} \right) = \frac{cN_{AV}}{M}$$

Polymers-Intrinsic Viscosity

Dilute Solutions ($c < c^$)*

Tend to investigate intrinsic viscosity

$$[\eta]_{\lim c \rightarrow 0} = \frac{\eta - \eta_s}{c\eta_s}$$

$$[\eta] = \Phi \frac{(\bar{r}^2)^{3/2}}{M};$$

$$\Phi = 2.6 \times 10^{23} \text{ mole}^{-1}$$

Flory Fox Equation

Intrinsic viscosity versus molecular weight

$$[\eta] \propto M^n \quad \text{What is } n?$$

Intrinsic viscosity – mol. wt.

$$[\eta] \propto M_w^n$$

$$[\eta] = \Phi \frac{(\bar{r}^2)^{3/2}}{M}$$

$$\bar{r}^2 \sim M$$

$$[\eta] \sim \frac{M^{3/2}}{M} \sim M^{1/2}$$

Experimentally:

$$[\eta] = K M^a$$

Mark Houwink relation

$$a = 0.5-0.8$$

$a \sim$ function of chain stiffness

Dilute Solutions

In the absence of long range interactions (theta conditions, one may relate the intrinsic viscosity to the chain dimensions as follows -

$$[\eta] = \Phi \left(\frac{\bar{r}_\theta^2}{M} \right)^{3/2} = CM^{1/2} = K_\theta M^{1/2}$$

Flory-Fox equation

K_θ is constant for a given polymer/solvent
independent of M

Φ = Flory viscosity constant
 $2.6 \times 10^{23} \text{ mol}^{-1}$

Mark-Houwink
relationship
for θ conditions

Polymers-Non-dilute

1. Follow analogy of suspensions:

$$\eta = \eta_s \left[1 + [\eta]c + k'[\eta]^2 c^2 + \dots \right]$$

$k'[\eta]^2 \equiv \text{Huggins Coefficient}$

k' is independent of molecular weight
is a function of solvent quality = 0.3-0.4 good solvent
= 0.5-0.8 θ solvent

2. Martin Equation:

$$\frac{\eta - \eta_s}{c\eta_s} = [\eta] \exp(kc[\eta])$$

If $k=k'$ then the two equations are similar

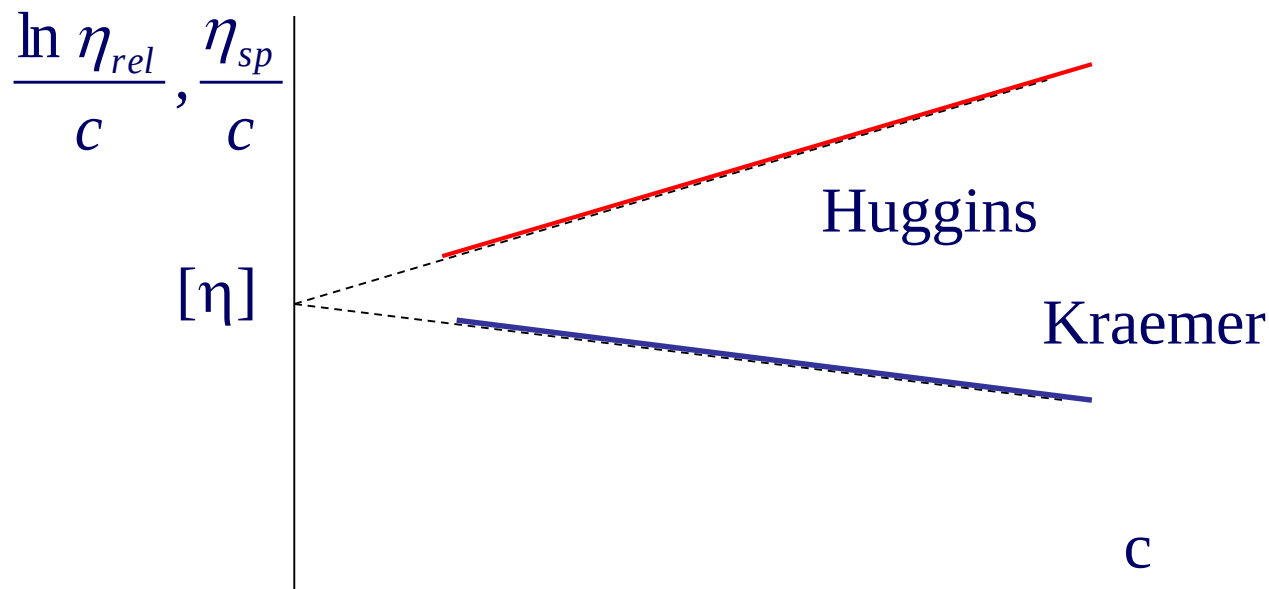
Solution Viscometry - Treatment of Data

Huggins Eqn: $\frac{\eta_{sp}}{c} = [\eta] + k'[\eta]^2 c$

Plot reduced viscosity (η_{sp}/c) vs. c
Intercept gives intrinsic viscosity
Slope = $k'[\eta]^2$

Kraemer Eqn: $\frac{\ln(\eta_r)}{c} = [\eta] + k''[\eta]^2 c$

Plot inherent viscosity $\ln(\eta_r)/c$ vs. c
Intercept gives intrinsic viscosity
Slope = $k''[\eta]^2$



Intrinsic Viscosity and the Mark-Houwink Relationship

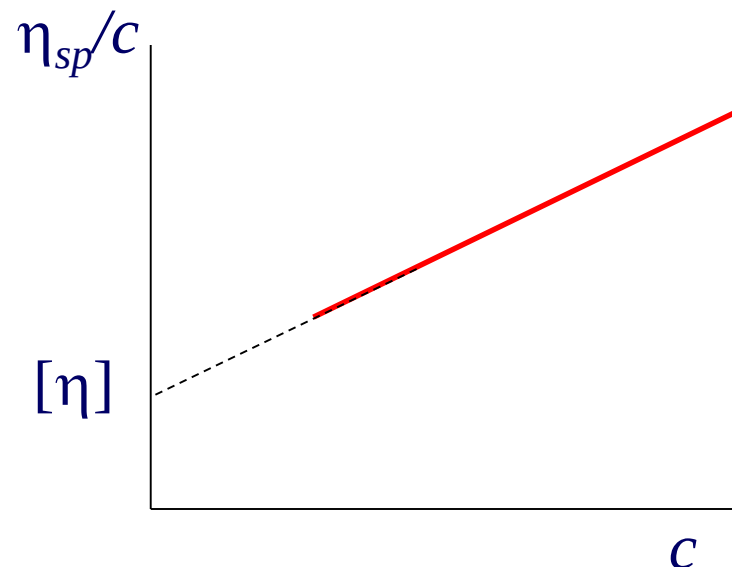
$$\frac{\eta}{\eta_s} = 1 + (c[\eta] + k'c^2[\eta]^2 + \dots)$$

Huggins Equation:

Coefficient k' relates to solvent quality

$k' \sim 0.3-0.4$ 'good' solvent

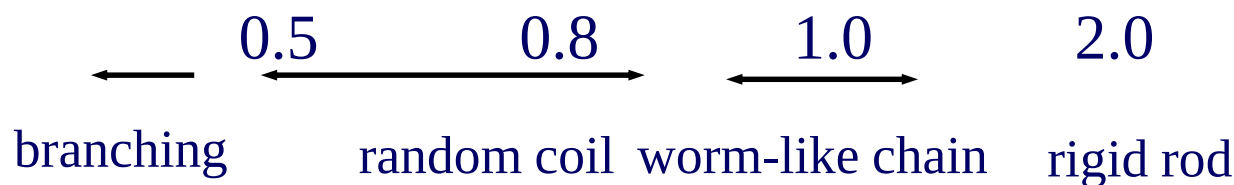
$k' \sim 0.5-0.8$ θ solvent



$$[\eta] = KM^\alpha$$

Mark - Houwink Equation:

Exponent α relates to coil “shape”



Some additional information

Dilute solution viscometry

“U-tube” viscometer:

$$\eta_r = \frac{\eta}{\eta_s} \approx \frac{t}{t_s}$$

Relative viscosity determined by comparison of efflux time of polymer solution and solvent

Concentration regimes

$$c^* \sim 1/[\eta]$$

Critical overlap concentration inversely related to intrinsic viscosity (and also chain size)

$$c^\# \sim 10c^*$$

The onset of chain entanglement occurs approximately at a polymer concentration in excess of $10c^*$

Polymers - Molecular Models

Kinetic theory; Physically based

- Dilute Solutions
- Concentrated Solutions or Melts

Dilute Solution

Elastic Dumbbell Model



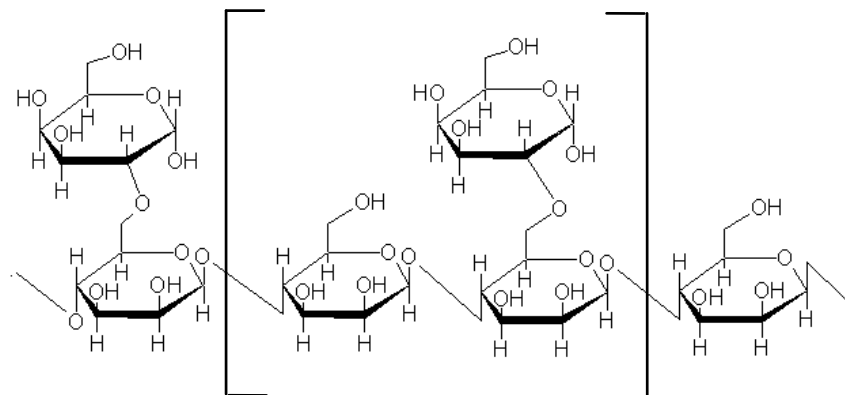
FENE Model



Rouse Model

Example Problem

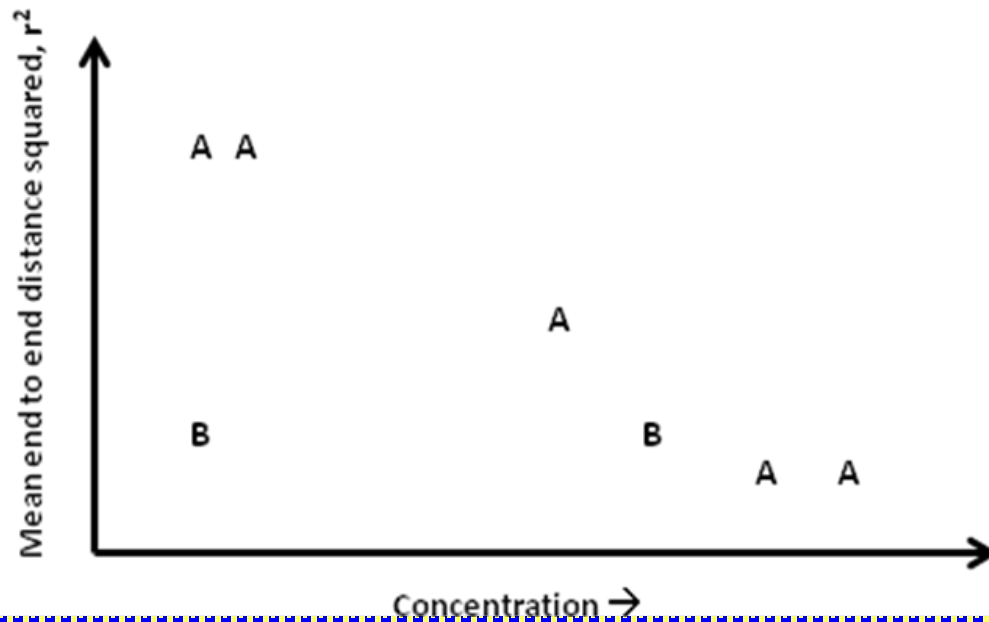
Consider the polymer below ($MW > 10^6$) with one side group sticking out for every two repeat units of the backbone. There are two possible enzymes, one that breaks the side groups and the other that cleaves the backbone randomly. (a) Your friend Joe claims that using the back-bone cleaving enzyme would have the most effect on viscosity. Does this make sense. Justify your answer. (b) Your other friend Bloe has used the side chain-cleaving enzyme to cleave some side groups. He comes back after a 7-day vacation and measures the viscosity of the enzymatically-modified sample and finds its viscosity higher than the original sample. Explain if this result is possible and why, or if your friend is wrong and why.



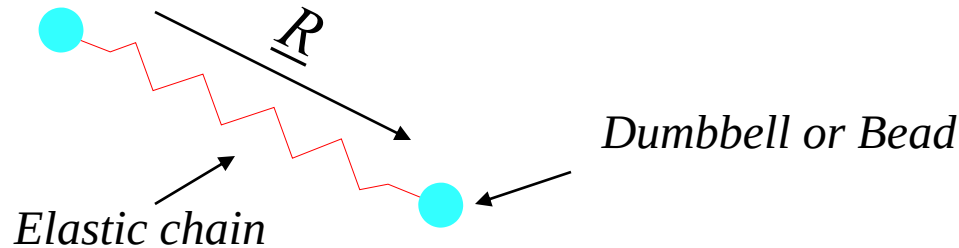
Example Problem

You are required to make a dilute solution of a polymer X in a good solvent (say toluene). How would you determine the maximumⁿ concentration (c^) of polymer you could use and still have a dilute polymer solution?*

You have two polymers A and B and one unknown solvent. Your dilemma is to determine if this is a good or theta solvent for each of your polymers. Based on the data obtained below, explain whether you think this is a good or theta solvent for each of your polymers. If the data is inconclusive, say so as well.

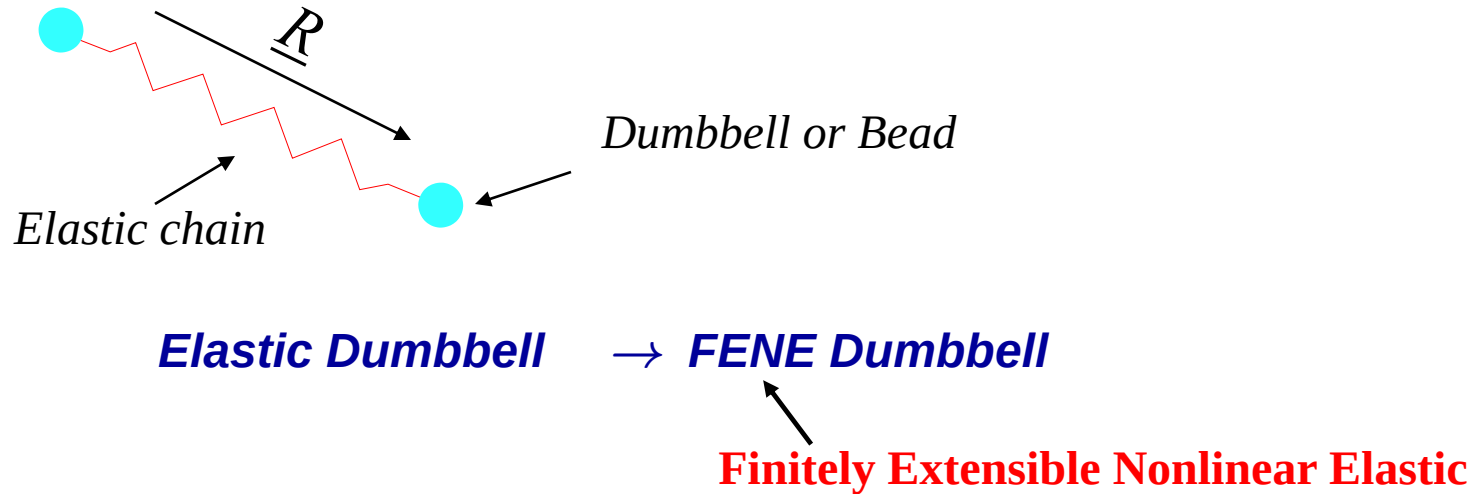


Elastic Dumbbell Model



- *Elastic spring can be extended infinitely*
 - *Beads act as points where viscous drag occurs*
 - *Elasticity comes from spring*
 - *Dumbbells account for concentration and molecular weight*
-
- **Looking at a single chain**
 - **Obtain Newtonian viscosity**

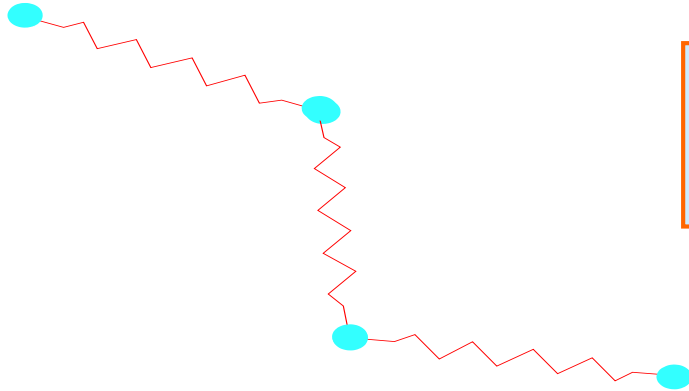
FENE Model



- Finite extension of spring
- Non-linear elasticity

These features start to bring in shear thinning behavior

Rouse Model



- Finite spring extension
- Freely jointed

- Can have N beads with $N-1$ springs
- Gives shear thinning
- Gives multiple relaxation times (λ_i)

$(\lambda_i) \sim N^2$ for large N

Rouse Model

At low ω

$$G' \sim \omega^2$$

$$G'' \sim \omega$$

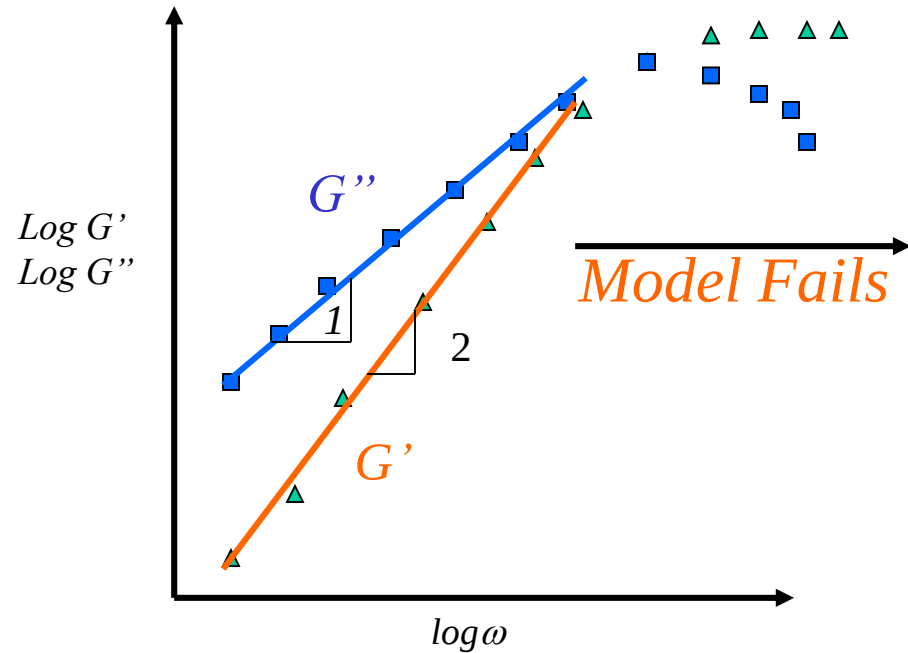
(expected behavior)

At high ω

$$G' \sim \omega^{1/2}$$

$$G'' \sim \omega^{1/2}$$

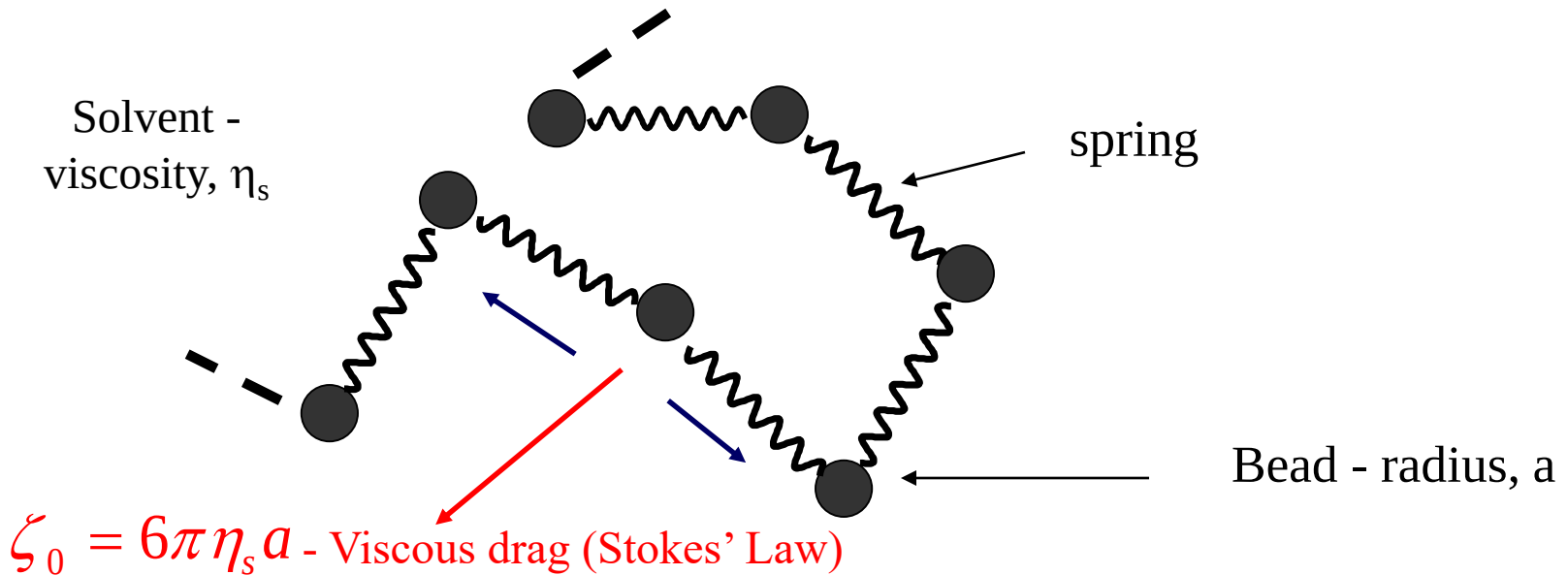
(Model fails)



At low ω Rouse model works for both dil. and conc. solutions & melt

Bead - Spring Theories (Rouse / Zimm)

- Theories developed independently - Rouse (1953), Zimm (1956)
- Valid for 'dilute' solutions of linear, flexible coils.
- Chains modelled as a series of $(N + 1)$ beads which represent the points of frictional interaction with the solvent.
- The beads are linked by N Hookean springs



Rouse vs Zimm Theory

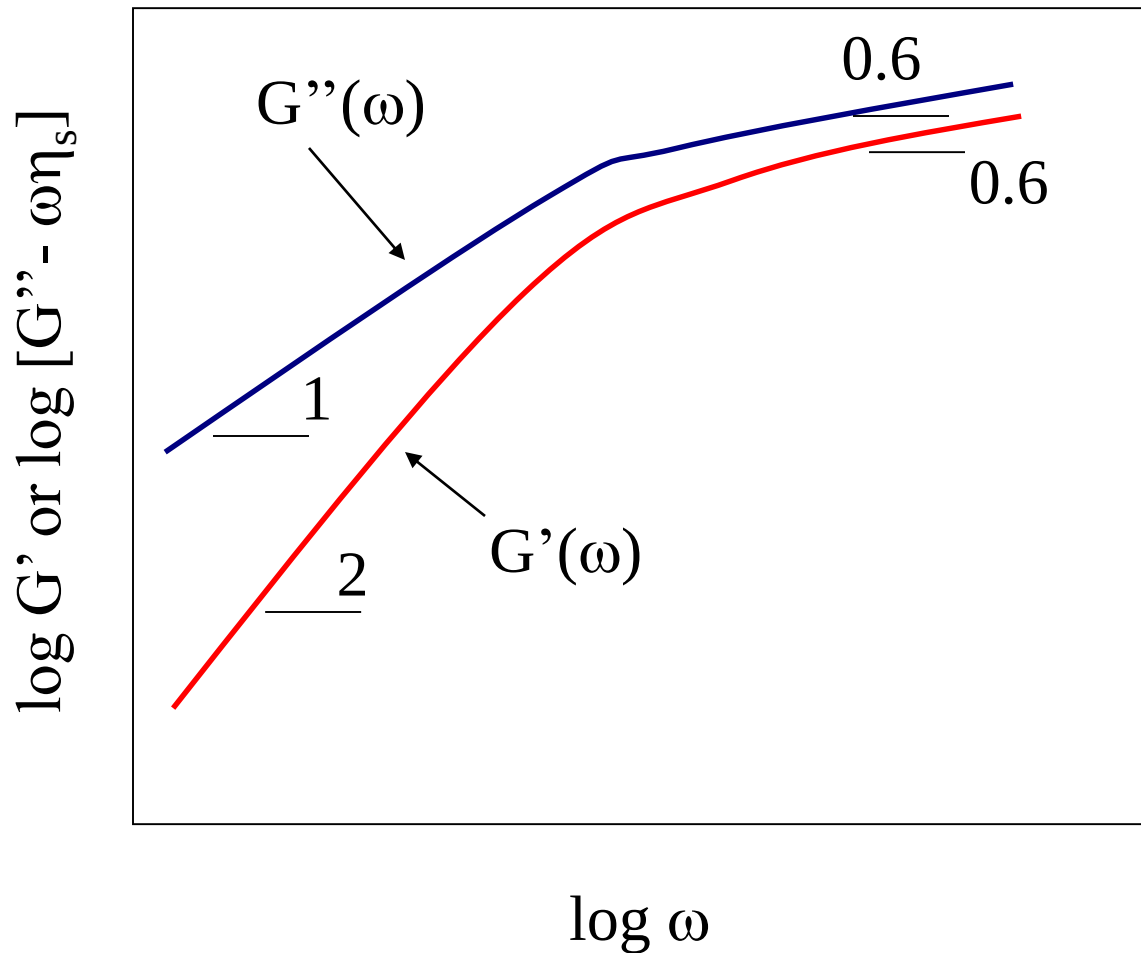
Rouse

- “Free Draining Chain”
- *Hydrodynamic interactions* caused by the motion of neighbouring beads along the chain are *ignored*.

Zimm

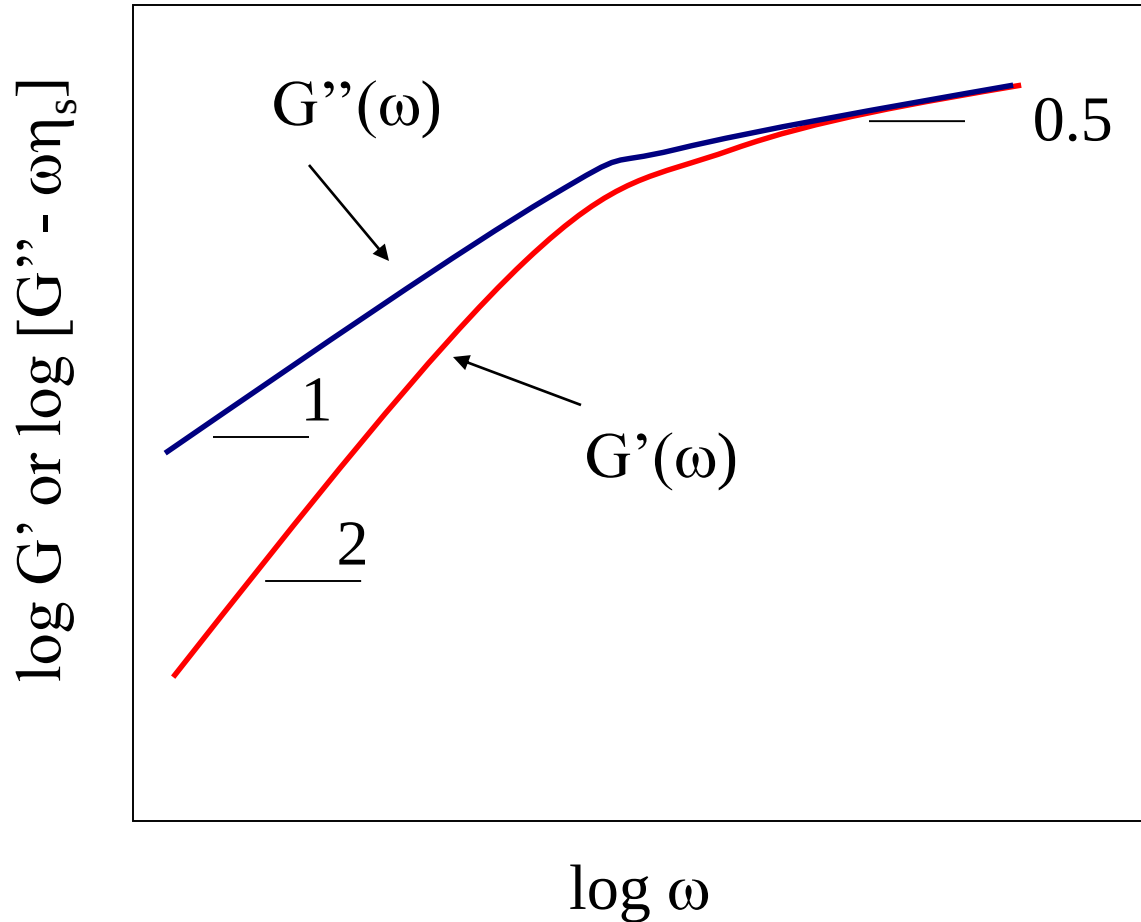
- “Non-Free Draining Chain”
- Considers frictional resistance to the motion of the beads in the bead-spring chain *and* viscous drag from beads in the same chain - “*dominant hydrodynamic interaction*”.
- Force on each bead is the sum of hydrodynamic drag force arising from its interaction with the solvent *plus* the Hookean spring forces exerted by the motion of the 2 neighbouring beads.

Zimm Model - Oscillatory Shear



- High frequency : $G'' = \sqrt{3} G'$

Rouse Model - Oscillatory Shear



- High frequency : $G' = G''$

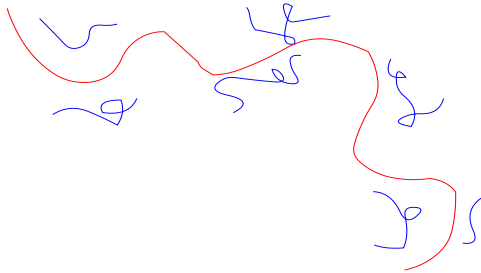
Frequency Dependence of the Dynamic Moduli

- Both Rouse and Zimm models show a “terminal region” at time-scales longer than λ_1 in which $G' \sim \omega^2$ and $G'' \sim \omega$.
- At higher frequencies the response is different as a consequence of the different spacings of the relaxation modes.
- Only the polymer chains contribute to elastic storage of energy as manifested in the magnitude of G' . Both chains and solvent contribute to viscous energy dissipation. Rouse model : G' and G'' scale with $\sim \omega^{0.5}$ and are equal in magnitude.
- Zimm model : G' and G'' scale with $\sim \omega^{0.6}$ and the viscous component is greater by a factor of $\sqrt{3}$

Concentrated Solutions or Melt

Reptation model

- *Doi & Edwards*
- *de Gennes*



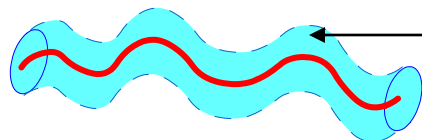
“Reptate” → snake-like motion

Focus in on a single chain

Surrounding chains

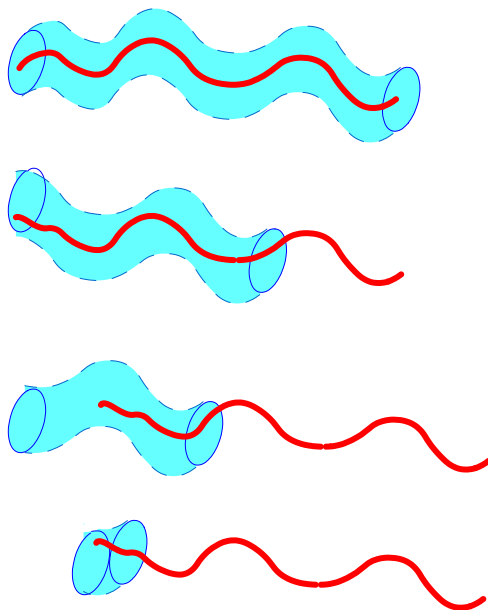
- **Prevent lateral motion**
- **Provide an equivalent tube around the chain**

Reptation Model



Equivalent tube formed by adjacent/surrounding polymers

Polymer chain can “wiggle” its way out of this imaginary tubes



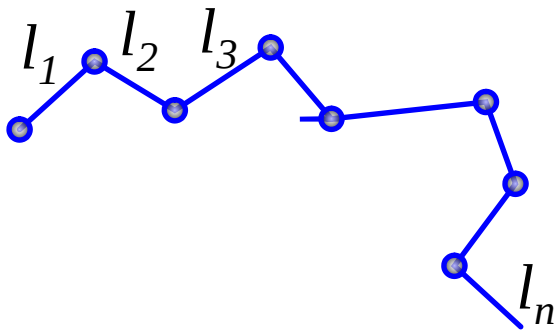
Reptation Model, cont'd

Diffusion coefficient is an important parameter

$$D_o \sim \frac{1}{M}$$

$$\lambda_{\text{reptation}} \sim \frac{L^2}{2D_o}$$

L = Statistical length of polymer chain



$$\mathbf{L=nl \sim M}$$

How is λ_{rep} related to mol. wt.?

$$\lambda_{\text{reptation}} \sim \frac{L^2}{2D_o} \sim \frac{M^2}{M^{-1}}$$

$$\lambda_{\text{reptation}} \sim M^3$$

Reptation Model, cont'd

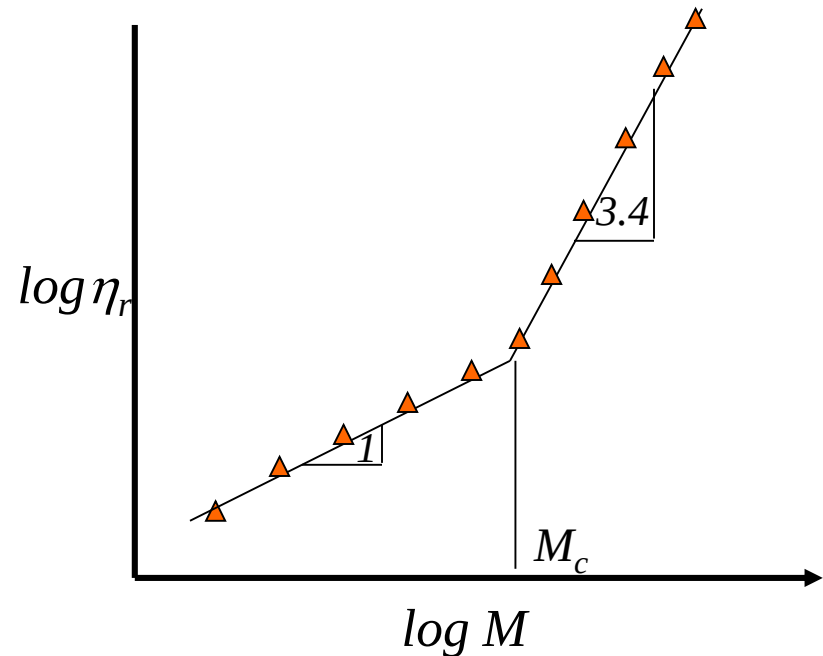
Experimental

$$\eta_{\text{reptation}} \sim M^{3.4}$$

Reptation model

$$\eta_{\text{reptation}} \sim M^3$$

How can we get this relation?



Reptation Model, cont'd

Reptation model

$$\eta_{\text{reptation}} \sim M^3$$

Suppose entire fluid behaves
as a simple viscoelastic fluid
(Maxwell element)

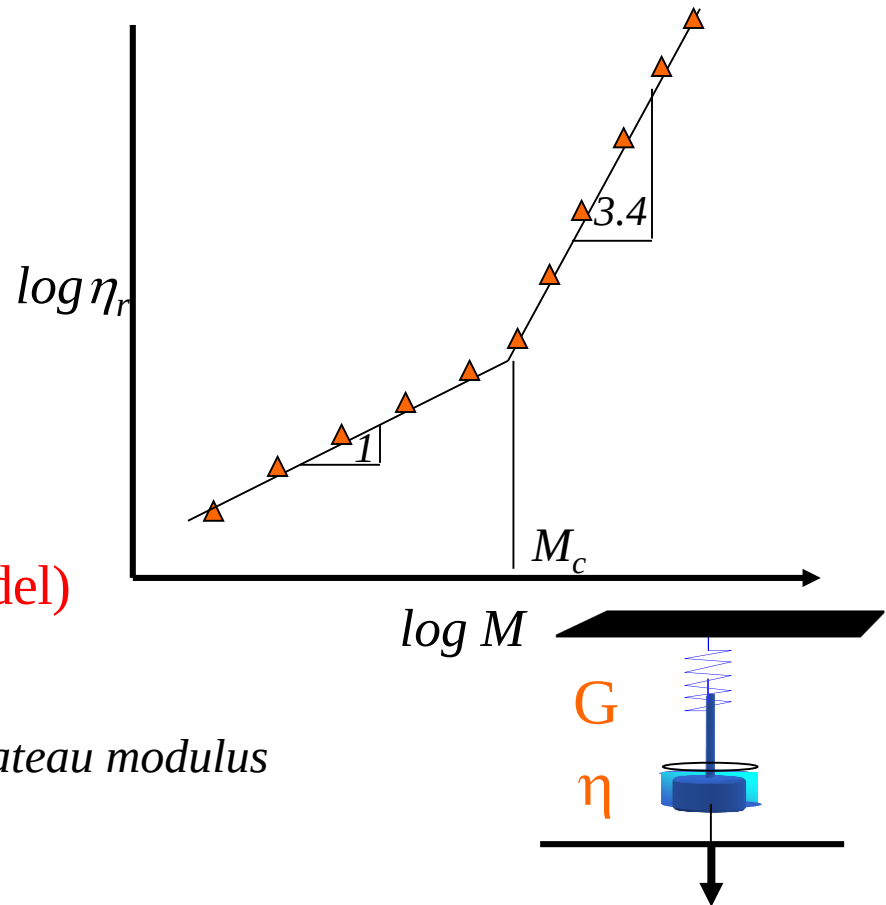
$$\lambda = \frac{\eta}{G} \quad (\text{Maxwell (LVE) Model})$$

$$\tau + \frac{\eta}{G} \dot{\tau} = \eta \dot{\gamma}$$

$$\eta \sim G \lambda \sim M^0 M^3$$

$$\eta \sim M^3$$

Reptation Model Prediction



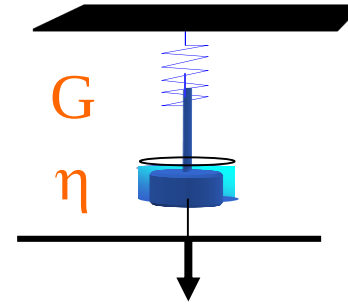
G =plateau modulus

Reptation Model, cont'd

Maxwell (LVE) Model

$$\underline{\underline{\tau}} = G(t) \gamma$$

relaxation modulus



$$\underline{\underline{\tau}} = G(t) h(\gamma) \gamma$$

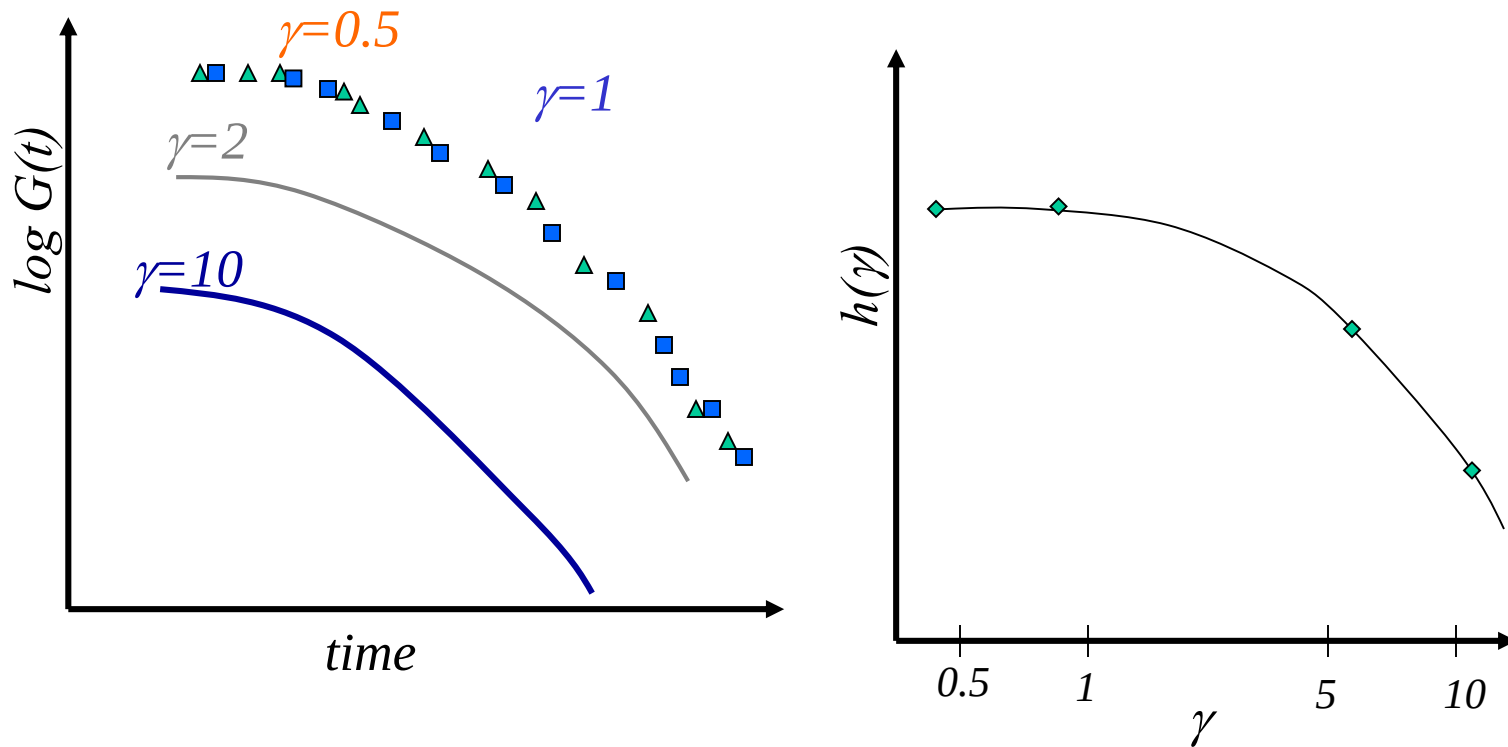
damping function

Doi-Edwards (Reptation Model)

- Damping function brings nonlinear features
- It is relevant in step strain (Stress-relaxation) experiments

Reptation Model, cont'd

Step-strain/stress relaxation experiments



Reptation Theory

- Entangled system - chains confined in a fictitious “tube” formed by a dense mesh of the surrounding chains → topological constraints.
- Tube is constrained at “entanglement” points and can only diffuse along the contour of the tube in a wormlike motion - *reptation*.
- The “tube” is continuously renewed as the chain emerges and explores laterally.
- Stress relaxation occurs via Rouse-like motion of strands between entanglement points and through reptation of the chains.

Reptation Theory

- The linear viscoelastic behaviour is given by:

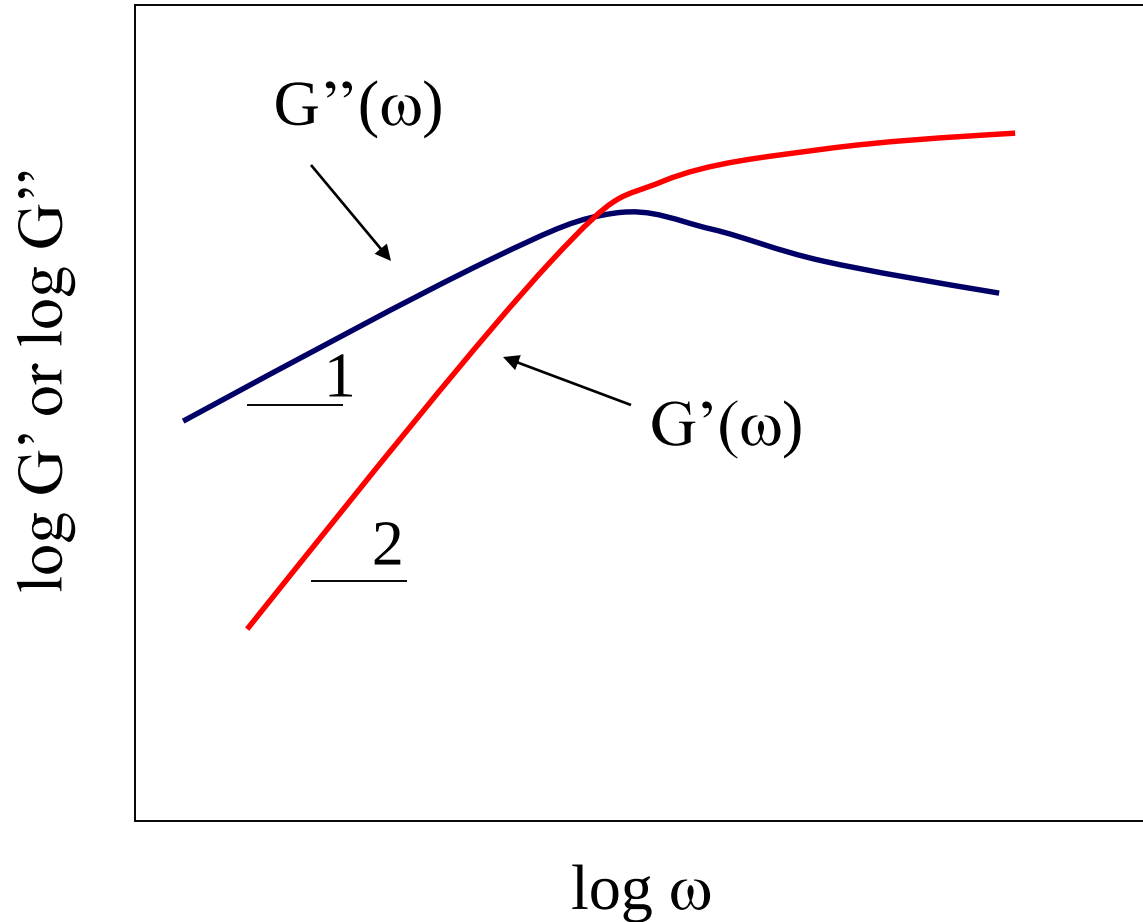
$$G'(\omega) = G_0 \sum_{k, \text{odd}} \frac{8}{\pi^2} \frac{1}{k^2} \frac{\omega^2 \lambda_k^2}{1 + \omega^2 \lambda_k^2}$$

$$G''(\omega) = G_0 \sum_{k, \text{odd}} \frac{8}{\pi^2} \frac{1}{k^2} \frac{\omega \lambda_k}{1 + \omega^2 \lambda_k^2}$$

- The spacing of relaxation modes is governed by:

$$\lambda_k = \frac{\lambda_1}{k^2} \quad k = 1, 3, 5, 7 \dots$$

Reptation Model - Oscillatory Shear



- High frequency : $G' > G''$